

PRISM: Personalized Retrieval via Implicit Subspace Mapping for Attribute-Aware Retrieval-Augmented Generation

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Abstract—Retrieval-Augmented Generation (RAG) has emerged as the dominant paradigm for grounding Large Language Models (LLMs) in external knowledge, yet existing approaches treat all users as a monolithic population, ignoring the heterogeneous information needs that arise in multi-domain enterprise environments. We formalize this deficiency as the *Context Pollution Problem* (CPP) and introduce PRISM (Personalized Retrieval via Implicit Subspace Mapping), a principled framework that integrates user-attribute-aware scoring into dense vector retrieval without retraining the embedding model. PRISM defines a *Dual-Space Attribute Alignment* (DSAA) mechanism that couples a semantic similarity score with a novel *Attribute Compatibility Kernel* (ACK), which operates over heterogeneous attribute spaces encompassing user roles and document metadata. We provide a Bayesian interpretation showing that PRISM’s scoring function approximates the posterior $P(d | q, u)$ under mild independence assumptions and prove that hard metadata filtering is a degenerate case of PRISM with strictly inferior recall. We further introduce *Cross-Domain Contamination Rate* (CDCR), a new evaluation metric for multi-domain retrieval purity. Extensive experiments on three benchmarks—ENTQA-5K, a role-stratified subset of MS MARCO, and BEIR—demonstrate that PRISM improves Domain Precision@5 by 43.8%, reduces CDCR by 62.1%, and achieves a 0.91 MRR on ambiguous multi-domain queries, while introducing only 0.3ms latency overhead per query. Ablation studies confirm the complementary contributions of role-category alignment and interest-tag matching, and scalability analysis shows sub-linear overhead up to 10^6 documents. Our work bridges Collaborative Filtering theory with RAG retrieval, opening a new research direction in *user-conditioned knowledge augmentation*.

I. INTRODUCTION

Large Language Models (LLMs) have achieved remarkable performance across a wide spectrum of natural language tasks [?], [?], [?]. Despite their extensive parametric knowledge, LLMs are prone to *hallucination*—generating plausible but factually incorrect statements—particularly when queried about domain-specific, time-sensitive, or proprietary information [?],

[?]. Retrieval-Augmented Generation (RAG) [?] addresses this by retrieving relevant passages from an external corpus $\mathcal{D}$ and conditioning the generative model on the retrieved context, thereby grounding outputs in verifiable sources.

a) Limitation of Existing RAG.: While RAG has proven effective in open-domain settings, a critical assumption pervades current architectures: *all users share the same relevance distribution*. Formally, given a query q , standard RAG retrieves documents by optimizing:

$$d^* = \arg \max_{d \in \mathcal{D}} \text{sim}(\phi(q), \phi(d)) \quad (1)$$

where $\phi : \mathcal{T} \rightarrow \mathbb{R}^h$ is a pre-trained text encoder and $\text{sim}(\cdot, \cdot)$ is typically cosine similarity. This formulation is *user-agnostic*: the identical query “What is the compliance policy?” issued by a Financial Analyst and an HR Manager yields identical retrieval results, despite the two users expecting fundamentally different documents. In enterprises with diverse organizational roles, this leads to what we term the **Context Pollution Problem (CPP)**—the contamination of retrieved context with semantically similar but domain-irrelevant documents.

b) Motivating Example.: Consider an enterprise knowledge base containing documents across Finance, HR, and Engineering domains. A Financial Analyst querying “annual review” expects documents about fiscal year financial reviews, while an HR Manager expects performance review templates. Dense retrievers, trained to capture semantic similarity, will return a heterogeneous mix of both, forcing the downstream LLM to disambiguate conflicting contexts—a task at which current models frequently fail [?]. We illustrate this phenomenon quantitatively in Section IV, where standard RAG achieves only 0.41 Domain Precision@5 on ambiguous cross-domain queries.